

Equation 7 Derivation

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Derivation of (7) from the SDF

Start from the key FX restriction:

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}].$$

Substitute $\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_{t+1} - v_{t+1}$, express $X_{i,t+1}$ linearly in g_{t+1} plus stationary variables. Under joint normality, only a linear loading on g_t balances the term involving g_{t+1} . Solving yields:

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top \mathbf{z}_t + u_{i,t},$$

with $\mu_{i,r}$ absorbing constant drift and Jensen corrections and $\Gamma_i^\top \mathbf{z}_t + u_{i,t}$ denoting the stationary block. $\square$

Deriving the affine form in (25)

Start from the FX restriction

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}],$$

with $\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}$, $\xi_{t+1} = \lambda_r^\top g_{t+1} + v_{t+1}$, $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$, and $v_{t+1} \perp \varepsilon_{t+1}$, jointly normal. Conjecture (to be verified) that $X_{i,t} = A_{i,r} + B_{i,r}^\top g_t + \Gamma_i^\top \mathbf{z}_t + u_{i,t}$ with $\mathbf{z}_t, u_{i,t}$ stationary and independent of ε_{t+1} . Then

$$\log(M_{t+1} e^{X_{i,t} - X_{i,t+1}}) = -r_t - \frac{1}{2}\varsigma_r^2 + B_{i,r}^\top g_t - (B_{i,r} + \lambda_r)^\top g_{t+1} - v_{t+1} - \Gamma_i^\top (\mathbf{z}_{t+1} - \mathbf{z}_t) - (u_{i,t+1} - u_{i,t}).$$

Since the shocks are jointly normal and enter linearly, the conditional expectation is the exponential of the conditional mean plus one half the conditional variance. Writing $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$, the deterministic (state-dependent) part contains $B_{i,r}^\top g_t - (B_{i,r} + \lambda_r)^\top (g_t + \mu_r) = -\lambda_r^\top g_t - (B_{i,r} + \lambda_r)^\top \mu_r$. For the restriction to hold for all states $(g_t, \mathbf{z}_t)$, the coefficient on g_t must vanish. Hence $B_{i,r} = \lambda_{i,r}$. All remaining constants and stationary terms are absorbed into $\mu_{i,r}$, Γ_i , and $u_{i,t}$, yielding the affine solution $X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top \mathbf{z}_t + u_{i,t}$. $\square$

Proposition (Affine representation of $X_{i,t}$)

Under regime r , suppose (i) the SDF is log-normal,

$$\log M_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \xi_{t+1}, \quad \xi_{t+1} = \lambda_r^\top g_{t+1} + v_{t+1},$$

with $\varsigma_r^2 = \lambda_r^\top \Sigma_{g,r} \lambda_r + \sigma_v^2$; (ii) the global factor follows a random walk $g_{t+1} = g_t + \mu_r + \sigma_{g,r}\varepsilon_{t+1}$; (iii) ε_{t+1} and v_{t+1} are conditionally Gaussian and independent; (iv) the $K \times 1$ fundamentals $\mathbf{z}_t$ are stationary and independent of ε_{t+1} , and the stationary block $\Gamma_i^\top \mathbf{z}_t + u_{i,t}$ has conditionally mean-zero first differences and finite variance.¹ Then the no-arbitrage FX restriction

$$1 = \mathbb{E}_t [M_{t+1} e^{X_{i,t} - X_{i,t+1}}] \tag{1}$$

admits a log-linear (affine) solution

$$X_{i,t} = \mu_{i,r} + \lambda_{i,r}^\top g_t + \Gamma_i^\top \mathbf{z}_t + u_{i,t}, \quad u_{i,t} \text{ stationary}, \tag{2}$$

¹Any nonzero *constant* drift in the stationary block can be absorbed into the intercept below.

with $\lambda_{i,r}$ a currency- and regime-specific loading on the single nonstationary factor g_t , and

$$\mu_{i,r} = -\bar{r}_r - \frac{1}{2}\varsigma_r^2 - (\lambda_{i,r} + \lambda_r)^\top \mu_r + \frac{1}{2}[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}(\Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t}))], \quad (3)$$

where $\bar{r}_r \equiv \mathbb{E}[r_t \mid r]$. Moreover, if $X_{i,t}$ had any additional I(1) component orthogonal to g_t , (1) would depend on that state and fail to hold for all $(g_t, \mathbf{z}_t)$; hence g_t spans the only nonstationary direction. Guess (2). Then

$$X_{i,t} - X_{i,t+1} = -\lambda_{i,r}^\top (g_{t+1} - g_t) - \Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t}) = -\lambda_{i,r}^\top \mu_r - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t}).$$

Stack the innovations $\eta_{t+1} \equiv (\varepsilon_{t+1}, v_{t+1}, \Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t}))'$. By assumptions, $\eta_{t+1} \mid \mathcal{F}_t$ is (jointly) normal with mean zero and finite covariance, and $Y_{t+1} \equiv \log M_{t+1} + X_{i,t} - X_{i,t+1}$ is *linear* in η_{t+1} :

$$Y_{t+1} = -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_{t+1} + \lambda_{i,r}^\top g_t - \lambda_{i,r}^\top \mu_r - v_{t+1} - \lambda_{i,r}^\top \sigma_{g,r} \varepsilon_{t+1} - \Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t}).$$

Using $g_{t+1} = g_t + \mu_r + \sigma_{g,r} \varepsilon_{t+1}$ gives

$$Y_{t+1} = \underbrace{-r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r}_{\text{deterministic given } \mathcal{F}_t} + \underbrace{-(\lambda_{i,r} + \lambda_r)^\top \sigma_{g,r} \varepsilon_{t+1} - v_{t+1} - \Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t})}_{\text{linear in Gaussian shocks}}.$$

Since $Y_{t+1} \mid \mathcal{F}_t$ is normal,

$$\mathbb{E}_t[e^{Y_{t+1}}] = \exp\left(\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1})\right).$$

Imposing (1) therefore requires

$$\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1}) = 0.$$

By the display above,

$$\begin{aligned} \mathbb{E}_t[Y_{t+1}] &= -r_t - \frac{1}{2}\varsigma_r^2 - \lambda_r^\top g_t - (\lambda_{i,r} + \lambda_r)^\top \mu_r, \\ \frac{1}{2} \text{Var}_t(Y_{t+1}) &= \frac{1}{2}[(\lambda_{i,r} + \lambda_r)^\top \Sigma_{g,r}(\lambda_{i,r} + \lambda_r) + \sigma_v^2 + \text{Var}_t(\Delta(\Gamma_i^\top \mathbf{z}_t + u_{i,t}))]. \end{aligned}$$

For $\mathbb{E}_t[Y_{t+1}] + \frac{1}{2} \text{Var}_t(Y_{t+1})$ to be *state-independent* (hold for all $(g_t, \mathbf{z}_t)$), any g_t -dependence in r_t must cancel the $-\lambda_r^\top g_t$ term; this is the standard affine restriction on short rates.² Taking expectations across r_t within the regime and collecting constants then yields (3). Finally, if $X_{i,t}$ contained any I(1) direction orthogonal to g_t , it would enter Y_{t+1} without a matching term in ξ_{t+1} , making $\mathbb{E}_t[e^{Y_{t+1}}]$ depend on that state and violating (1). Hence g_t is the *only* nonstationary driver, and (2) follows.

²Allow the short rate to be affine in the state, $r_t = r_{0,r} + r_{g,r}^\top g_t + r_{z,r}^\top \mathbf{z}_t$; the identity for all g_t implies $r_{g,r} = -\lambda_r$ and $r_{z,r} = 0$. This matches the “prices are exponential-affine in states” logic of affine pricing.